S. B. JAIN INSTITUTE OF TECHNOLOGY, MANAGEMENT & RESEARCH, NAGPUR.

Practical No. 8

Aim: Demonstrate the concept of Semantic network on imputed predicate logic to represent the knowledge.

Name of Student:

Roll No.:

Semester/Year: VI/III

Academic Session:

Date of Performance:

Date of Submission:

AIM: Demonstrate the concept of Semantic network on imputed predicate logic to represent the knowledge.

OBJECTIVE/EXPECTED LEARNING OUTCOME:

The objectives and expected learning outcome of this practical are:

- Understand semantic networks as a knowledge representation technique
- Implement ISA and HAS-A relationships
- Demonstrate inheritance in semantic networks
- Query knowledge programmatically using Python

THEORY:

Semantic Network

A **semantic network** is a graphical knowledge representation technique consisting of:

- **Nodes** → Concepts / Objects
- **Edges** → Relationships

Common relations:

- ISA → Inheritance
- HAS-A → Property
- CAN → Ability

It is closely related to **predicate logic representation**, where:

isa(dog, animal).
has(dog, tail).
can(bird, fly).

Relationship to Predicate Logic

A semantic network can be represented using first-order predicates:

Semantic Relation	Predicate Form
Dog is an Animal	isa(dog, animal)
Bird can Fly	can(bird, fly)
Dog has Tail	has(dog, tail)

Inheritance rule:

If

isa(X, Y) and has(Y, Z)

Then

has(X, Z)

Semantic Network Example

Nodes:

- Animal
- Bird
- Dog
- Sparrow

Relationships:

- Bird ISA Animal
- Dog ISA Animal
- Sparrow ISA Bird
- Animal has cells
- Bird can fly
- Dog can bark

PROGRAM CODE:

```
# Semantic Network using Dictionary
def semantic_network():
    network = {}   # dictionary to store relations

    n = int(input("Enter number of relations: "))

    for i in range(n):
        print(f"\nEnter details for relation {i+1}:")
        subject = input("Enter Subject: ")
        relation = input("Enter Relation (e.g., ISA, HAS): ")
        obj = input("Enter Object: ")

        # Add to dictionary
        if subject not in network:
            network[subject] = []

        network[subject].append((relation, obj))

    print("\n--- Semantic Network Representation ---")

    # Display in required format
    for subject in network:
        for (relation, obj) in network[subject]:
            print(f"{subject} ---> {relation} ----> {obj}")

    return network

# Run program and get network data
network_data = semantic_network()

# Visualization code
import matplotlib.pyplot as plt
import networkx as nx

# Install networkx if not already installed
try:
    import networkx as nx
except ImportError:
    !pip install networkx
    import networkx as nx

# Install matplotlib if not already installed
try:
    import matplotlib.pyplot as plt
except ImportError:
    !pip install matplotlib
    import matplotlib.pyplot as plt
```

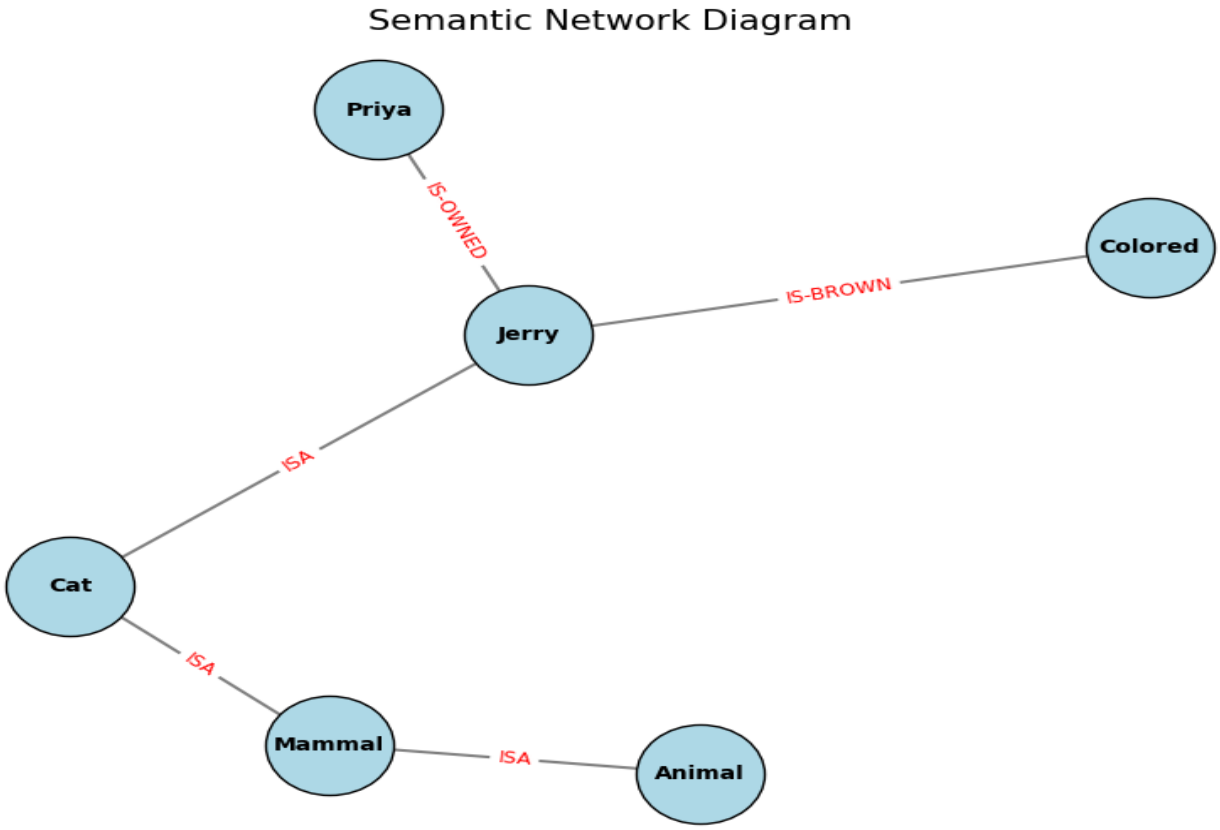
```
# Create a directed graph
G = nx.DiGraph()

# Add nodes and edges from the network data
for subject, relations in network_data.items():
    G.add_node(subject)
    for relation, obj in relations:
        G.add_node(obj)
        G.add_edge(subject, obj, label=relation)

# Draw the graph
plt.figure(figsize=(10, 8))
pos = nx.spring_layout(G, k=0.8) # Position nodes using Fruchterman-Reingold force-directed
algorithm

nx.draw_networkx_nodes(G, pos, node_size=3000, node_color='lightblue', edgecolors='black')
nx.draw_networkx_labels(G, pos, font_size=10, font_weight='bold')
nx.draw_networkx_edges(G, pos, edge_color='gray', arrowsize=20, width=1.5)

edge_labels = nx.get_edge_attributes(G, 'label')
nx.draw_networkx_edge_labels(G, pos, edge_labels=edge_labels, font_color='red', font_size=9)

plt.title('Semantic Network Diagram', size=15)
plt.axis('off') # Hide axes
plt.show()
```

INPUT & OUTPUT:

```
...  Enter number of relations: 5

     Enter details for relation 1:
     Enter Subject: Jerry
     Enter Relation (e.g., ISA, HAS): ISA
     Enter Object: Cat

     Enter details for relation 2:
     Enter Subject: Cat
     Enter Relation (e.g., ISA, HAS): ISA
     Enter Object: Mammal

     Enter details for relation 3:
     Enter Subject: Mammal
     Enter Relation (e.g., ISA, HAS): ISA
     Enter Object: Animal
```

```
Enter details for relation 4:
Enter Subject: Jerry
Enter Relation (e.g., ISA, HAS): IS-OWNED
Enter Object: Priya

Enter details for relation 5:
Enter Subject: Jerry
Enter Relation (e.g., ISA, HAS): IS-BROWN
Enter Object: Colored

--- Semantic Network Representation ---
Jerry ---> ISA ----> Cat
Jerry ---> IS-OWNED ----> Priya
Jerry ---> IS-BROWN ----> Colored
Cat ---> ISA ----> Mammal
Mammal ---> ISA ----> Animal
```

SEMENTIC NETWORK DIAGRAM:



CONCLUSION:

DISCUSSION QUESTIONS:

1. What is a semantic network?

2. What is ISA relationship?

3. What is inheritance in knowledge representation?

4. How is predicate logic used here?

5. What are advantages of semantic networks?

REFERENCES:

- *https://www.geeksforgeeks.org/artificial-intelligence/semantic-networks-in-artificial-intelligence/*
- *https://thecloudflare.com/knowledge-representation-using-the-semantic-network-in-ai/*
- *https://www.siliconvalleycloudit.com/knowledge-representation-in-ai-semantic-networks/*